

StructLab

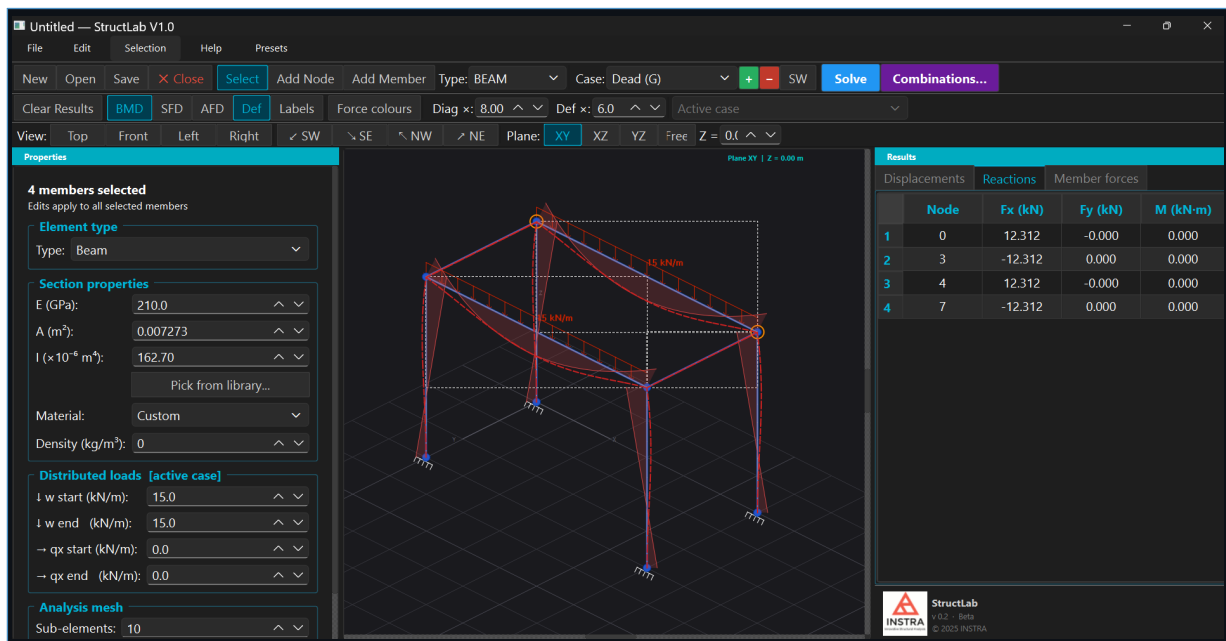
PHASE 8

Benchmark & Validation Report

StructLab vs OpenSeesPy · PyNite · Analytical

Linear-Elastic Static Analysis · Direct Stiffness Method

2D Beams (7) · 2D Frames (3) · 2D Trusses (2) · 3D Frames (8)

20/20 Cases PASS · 72/72 Quantities $\leq 0.10\%$ Error

Generated 18 May 2026 · Python 3.12 · PyQt6

Executive Summary

This report presents the results of 20 independent benchmark cases validating the StructLab Direct Stiffness Method engine against OpenSeesPy, PyNiteFEA, and closed-form analytical solutions. All 72 compared quantities must have a relative error $\leq 0.10\%$ to be marked PASS.

ID	Title	Category	Reference	Q	Max err	Status
B1	Simply Supported Beam — Mid-span Point Load	2D Beams	analytical + OpenSeesPy	5	0.0000%	ALL PASS
B2	Simply Supported Beam — UDL	2D Beams	analytical + OpenSeesPy	5	0.0000%	ALL PASS
B3	Propped Cantilever — Mid-span Point Load	2D Beams	analytical + OpenSeesPy	5	0.0000%	ALL PASS
B4	Propped Cantilever — UDL	2D Beams	analytical + OpenSeesPy	4	0.0000%	ALL PASS
B5	2-span Continuous Beam — UDL	2D Beams	analytical + OpenSeesPy	5	0.0000%	ALL PASS
B6	3-span Continuous Beam — UDL	2D Beams	OpenSeesPy	4	0.0000%	ALL PASS
B7	Fixed-Fixed Beam — UDL	2D Beams	analytical + OpenSeesPy	5	0.0000%	ALL PASS
F1	Portal Frame — Lateral Point Load	2D Frames	OpenSeesPy	4	0.0000%	ALL PASS
F2	2-storey Portal Frame — Gravity UDL on Beams	2D Frames	OpenSeesPy	3	0.0000%	ALL PASS
F3	2-bay Portal Frame — Gravity UDL + Lateral Load	2D Frames	OpenSeesPy	4	0.0000%	ALL PASS
T1	Pratt Truss — Equal Panel Point Loads	2D Trusses	analytical + OpenSeesPy	4	0.0000%	ALL PASS
T2	Warren Truss — Mid-span Point Load	2D Trusses	analytical + OpenSeesPy	4	0.0000%	ALL PASS
3D1	3D Cantilever Column — Tip Point Load	3D Frames	analytical + PyNite	4	0.0000%	ALL PASS
3D2	3D Cantilever — Vertical Tip Load (Strong Axis)	3D Frames	analytical + PyNite	4	0.0000%	ALL PASS
3D3	3D Simply Supported Beam — UDL	3D Frames	analytical + PyNite	3	0.0000%	ALL PASS
3D4	2-storey Space Frame — Gravity Loads	3D Frames	PyNite	2	0.0000%	ALL PASS
3D5	L-shaped 3D Frame — Vertical Tip Load	3D Frames	PyNite	1	0.0365%	ALL PASS
3D6	3D Portal Frame — Gravity UDL + Lateral Load	3D Frames	PyNite	2	0.0000%	ALL PASS
3D7	Space Truss — Tetrahedral Apex, Vertical Point L	3D Frames	analytical + PyNite	2	0.0000%	ALL PASS
3D8	3-bay Single-storey 3D Frame — Gravity UDL	3D Frames	analytical + PyNite	2	0.0000%	ALL PASS

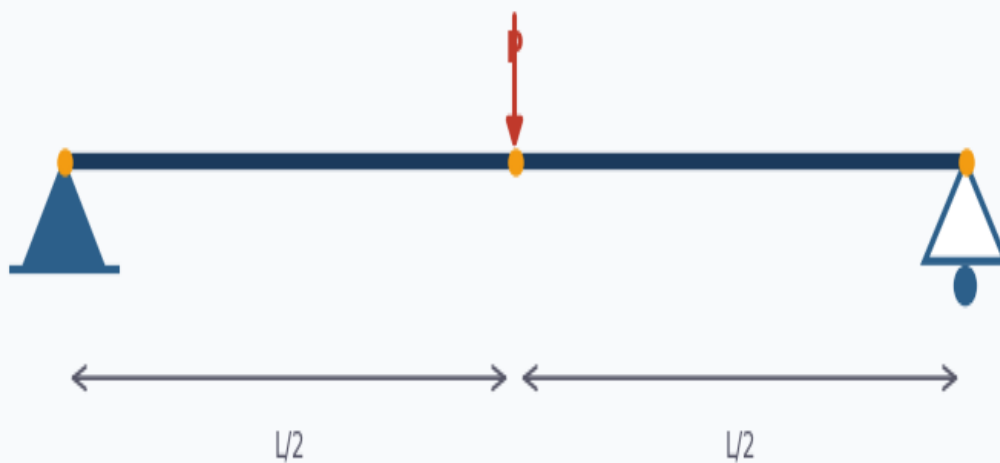
✓ 20/20 cases PASS — maximum error across all quantities: 0.0365%

2D Beams

B1 — Simply Supported Beam — Mid-span Point Load

Span $L=6.0$ m, point load $P=50$ kN at mid-span. $E=210$ GPa, $I=400$ cm⁴. Validated against analytical DSM solution and OpenSeesPy.

B1 — Simply Supported Beam, Mid-span Point Load

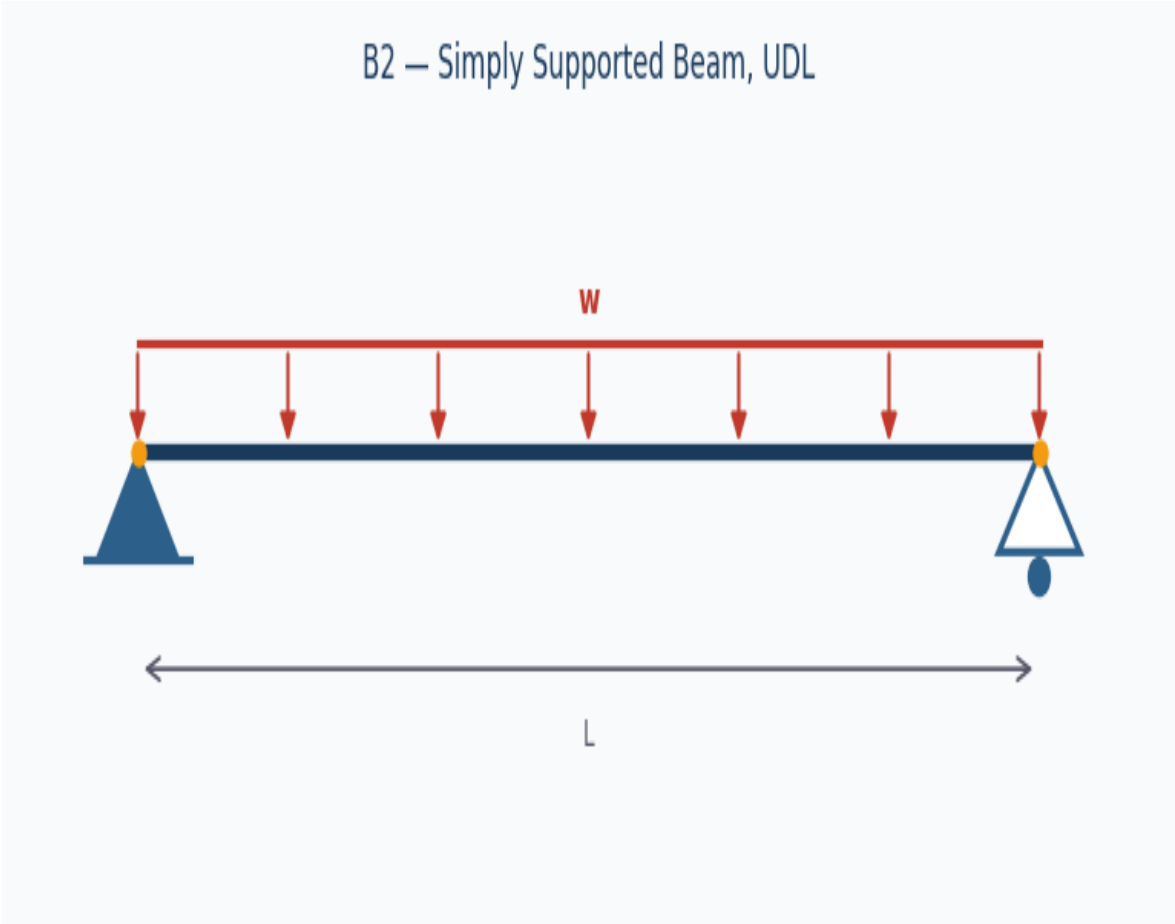


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Mid-span deflection (mm)	2.6786	2.6786	analytical	0.0000%	PASS
Mid-span deflection (mm)	2.6786	2.6786	OpenSeesPy	0.0000%	PASS
Reaction R_A (kN)	0.0250	0.0250	analytical	0.0000%	PASS
Reaction R_B (kN)	0.0250	0.0250	analytical	0.0000%	PASS
Max moment $M_{\max}$ (kN·m)	0.0750	0.0750	analytical	0.0000%	PASS

✓ ALL 5 QUANTITIES PASS — max error 0.0000%

B2 — Simply Supported Beam — UDL

Span $L=6.0$ m, UDL $w=20$ kN/m. $E=210$ GPa, $I=400$ cm⁴. Validated against analytical and OpenSeesPy.

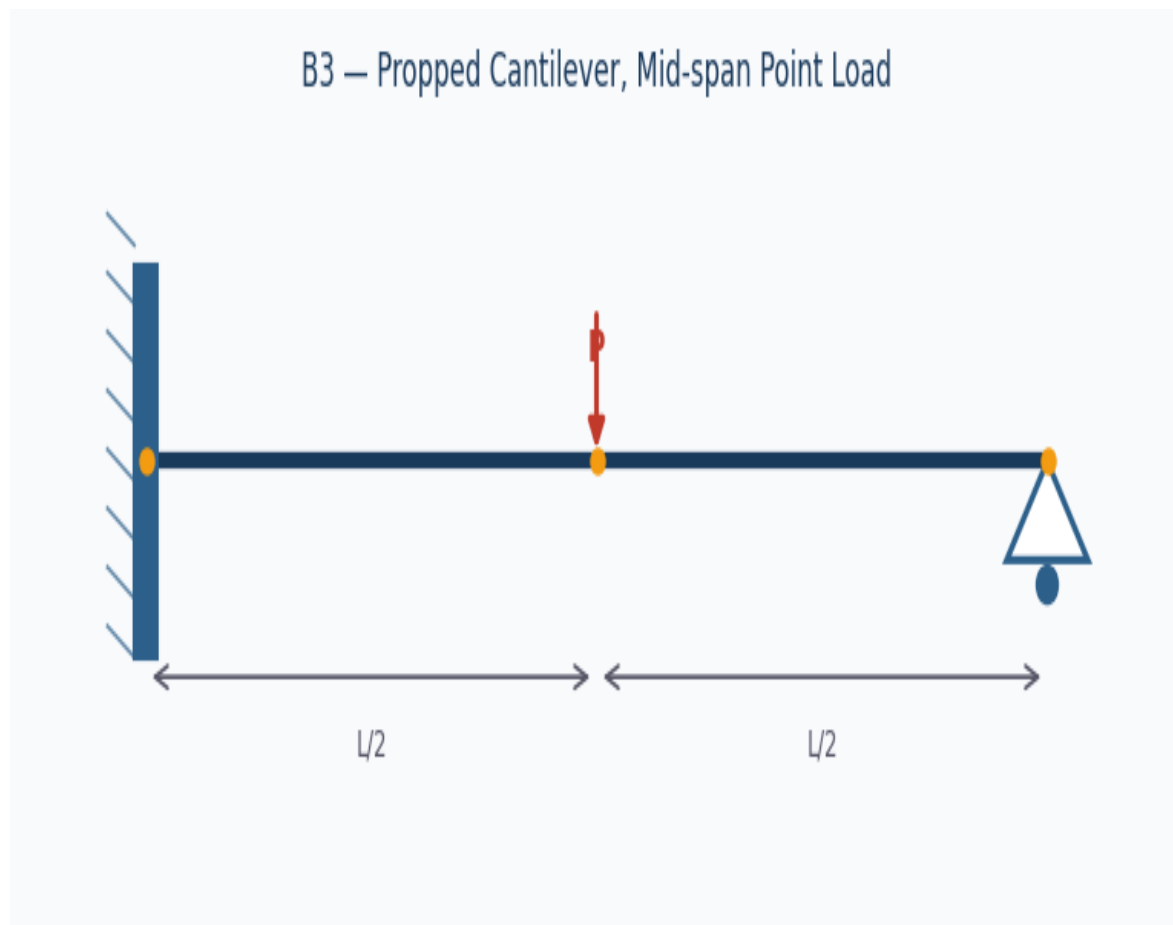


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Mid-span deflection (mm)	4.0179	4.0179	analytical	0.0000%	PASS
Mid-span deflection (mm)	4.0179	4.0179	OpenSeesPy	0.0000%	PASS
Reaction R_A (kN)	0.0600	0.0600	analytical	0.0000%	PASS
Reaction R_A (kN)	0.0600	0.0600	OpenSeesPy	0.0000%	PASS
Max moment M_max (kN·m)	0.0900	0.0900	analytical	0.0000%	PASS

✓ ALL 5 QUANTITIES PASS — max error 0.0000%

B3 — Propped Cantilever — Mid-span Point Load

$L=5.0$ m, fixed at A, roller at B, point load $P=40$ kN at mid-span. Validated against force-method analytical solution and OpenSeesPy.

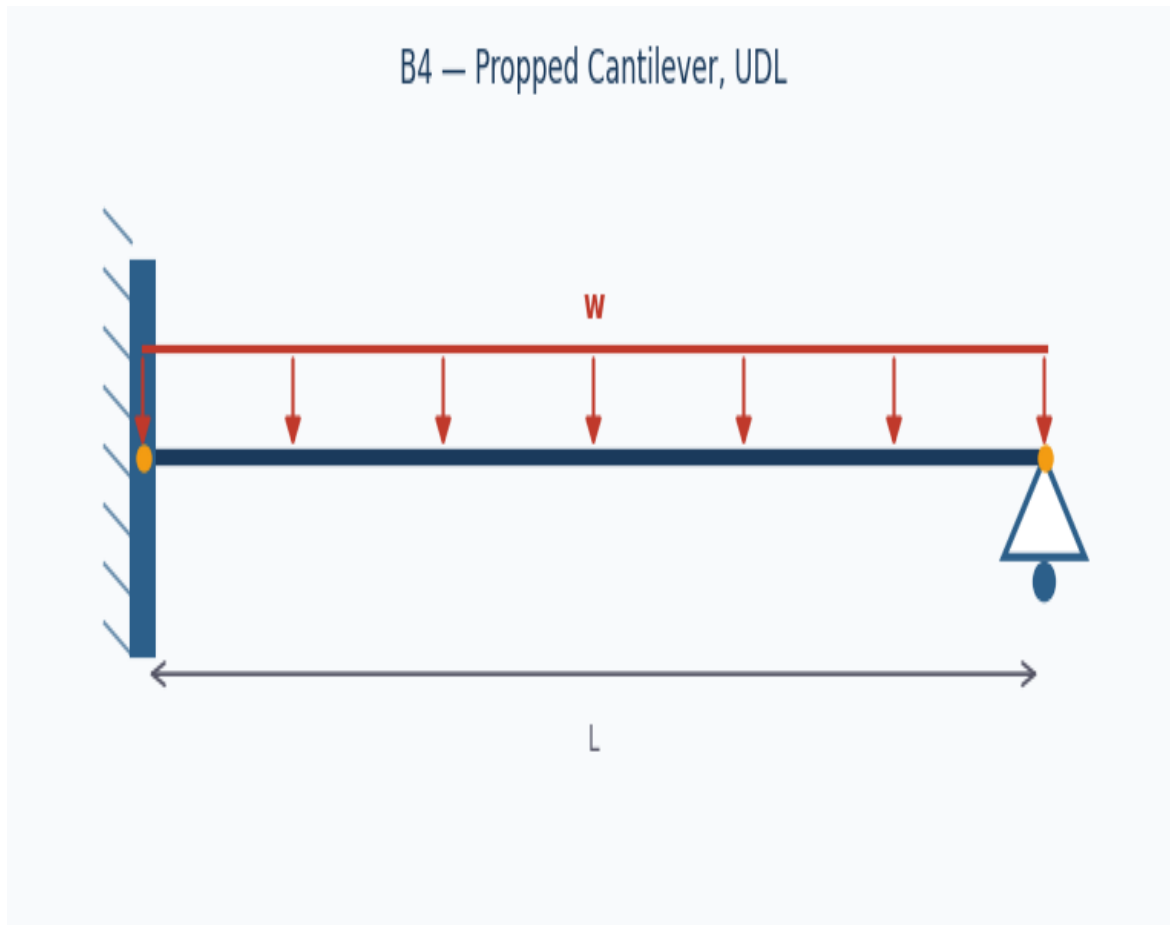


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Mid-span deflection (mm)	0.7595	0.7595	analytical	0.0000%	PASS
Mid-span deflection (mm)	0.7595	0.7595	OpenSeesPy	0.0000%	PASS
Roller reaction R_B (kN)	0.0125	0.0125	analytical	0.0000%	PASS
Root moment $ M_A $ (kN·m)	0.0375	0.0375	analytical	0.0000%	PASS
Root moment $ M_A $ (kN·m)	0.0375	0.0375	OpenSeesPy	0.0000%	PASS

✓ ALL 5 QUANTITIES PASS — max error 0.0000%

B4 — Propped Cantilever — UDL

$L=6.0$ m, fixed at A, roller at B, UDL $w=15$ kN/m. Standard force-method result: $R_B=3wL/8$, $M_A=wL^2/8$.

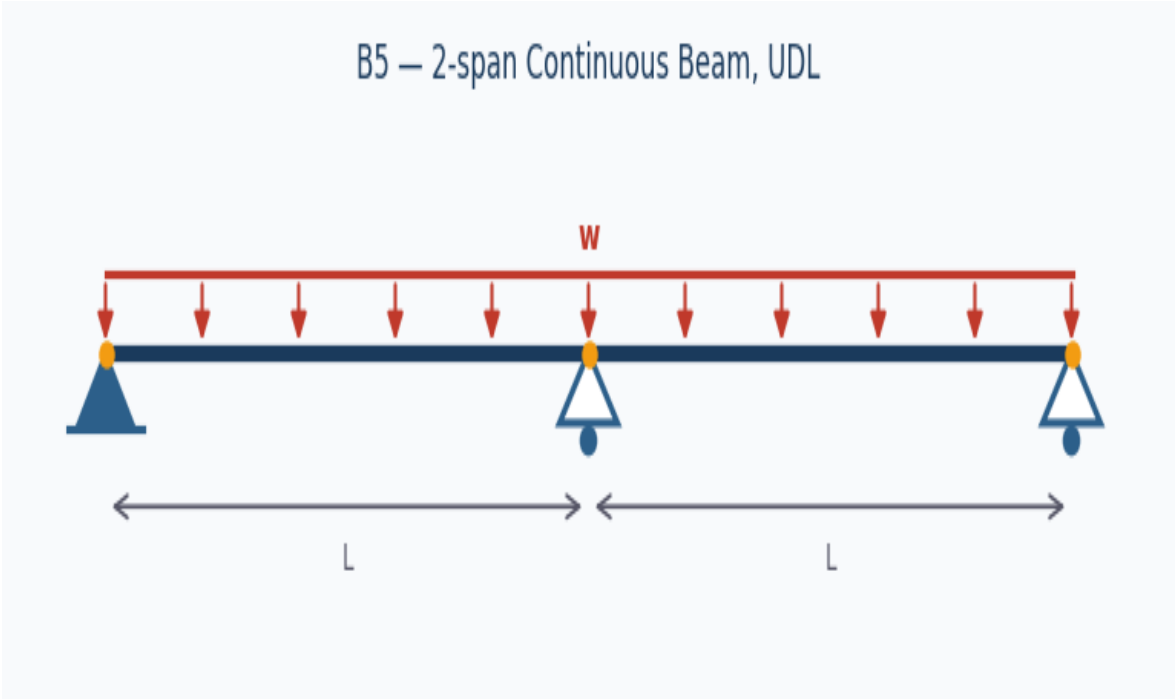


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Roller reaction R_B (kN)	0.0338	0.0338	analytical	0.0000%	PASS
Roller reaction R_B (kN)	0.0338	0.0338	OpenSeesPy	0.0000%	PASS
Root moment M_A (kN·m)	0.0675	0.0675	analytical	0.0000%	PASS
Root moment M_A (kN·m)	0.0675	0.0675	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

B5 — 2-span Continuous Beam — UDL

Two equal spans $L=5.0$ m, UDL $w=20$ kN/m on both. 3-moment theorem: end reactions = $3wL/8$, centre reaction = $5wL/4$.

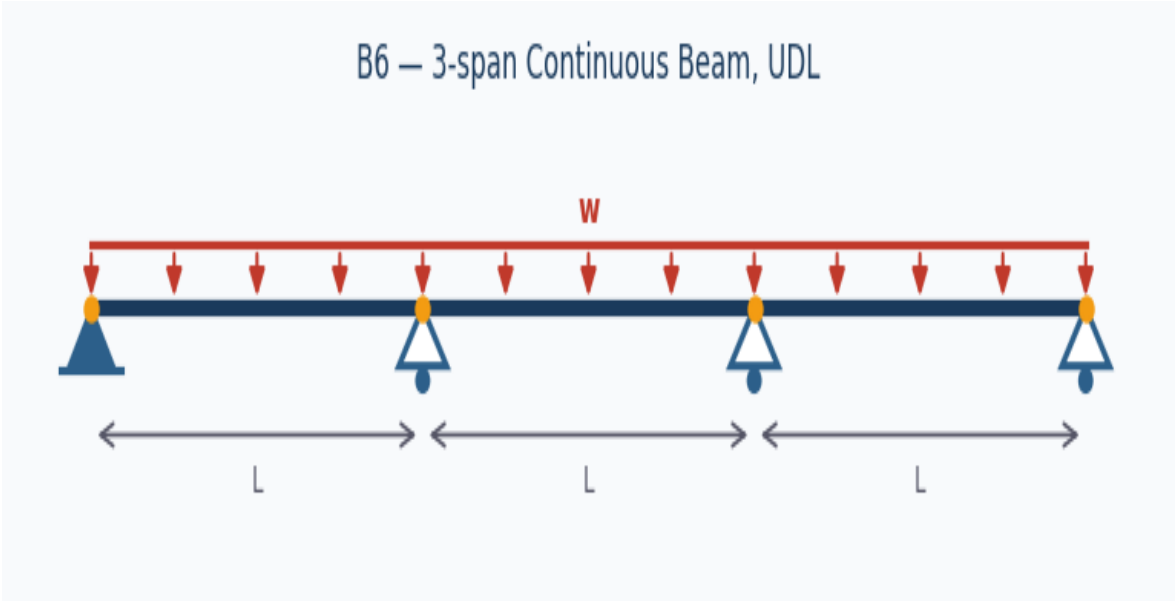


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Reaction R_A (end) (kN)	0.0375	0.0375	analytical	0.0000%	PASS
Reaction R_A (end) (kN)	0.0375	0.0375	OpenSeesPy	0.0000%	PASS
Reaction R_mid (kN)	0.1250	0.1250	analytical	0.0000%	PASS
Reaction R_mid (kN)	0.1250	0.1250	OpenSeesPy	0.0000%	PASS
Reaction R_B (end) (kN)	0.0375	0.0375	analytical	0.0000%	PASS

✓ ALL 5 QUANTITIES PASS — max error 0.0000%

B6 — 3-span Continuous Beam — UDL

Three equal spans $L=4.0$ m, UDL $w=25$ kN/m. Statically indeterminate to 2nd degree. Reactions validated vs OpenSeesPy.

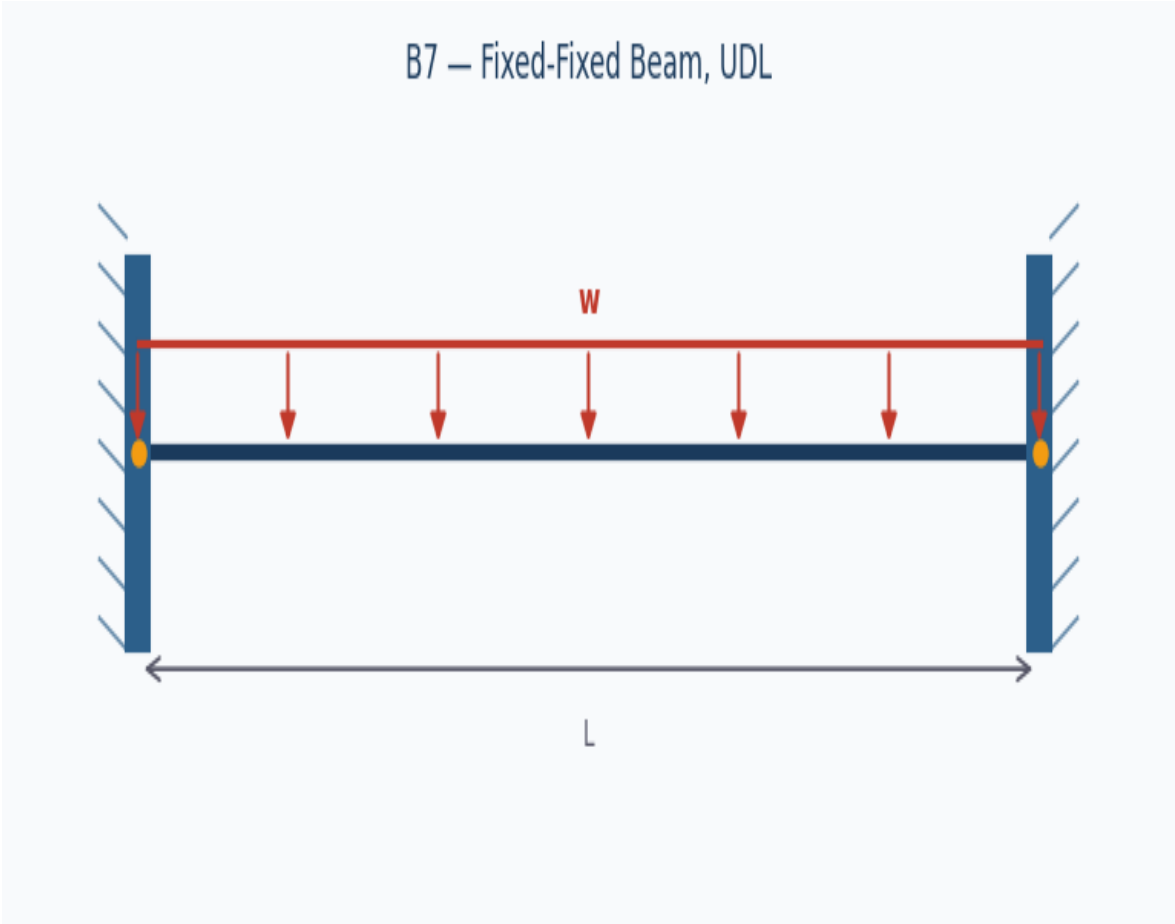


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Reaction R■ (kN)	0.0187	0.0188	OpenSeesPy	0.0000%	PASS
Reaction R■ (kN)	-0.0000	0.0000	OpenSeesPy	0.0000%	PASS
Reaction R■ (kN)	0.1312	0.1312	OpenSeesPy	0.0000%	PASS
Reaction R■ (kN)	0.1500	0.1500	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

B7 — Fixed-Fixed Beam — UDL

Span $L=5.0$ m, both ends fully fixed, UDL $w=30$ kN/m. Analytical: $M_{ends}=wL^2/12$, $M_{mid}=wL^2/24$, $\delta_{max}=wL^4/384EI$.



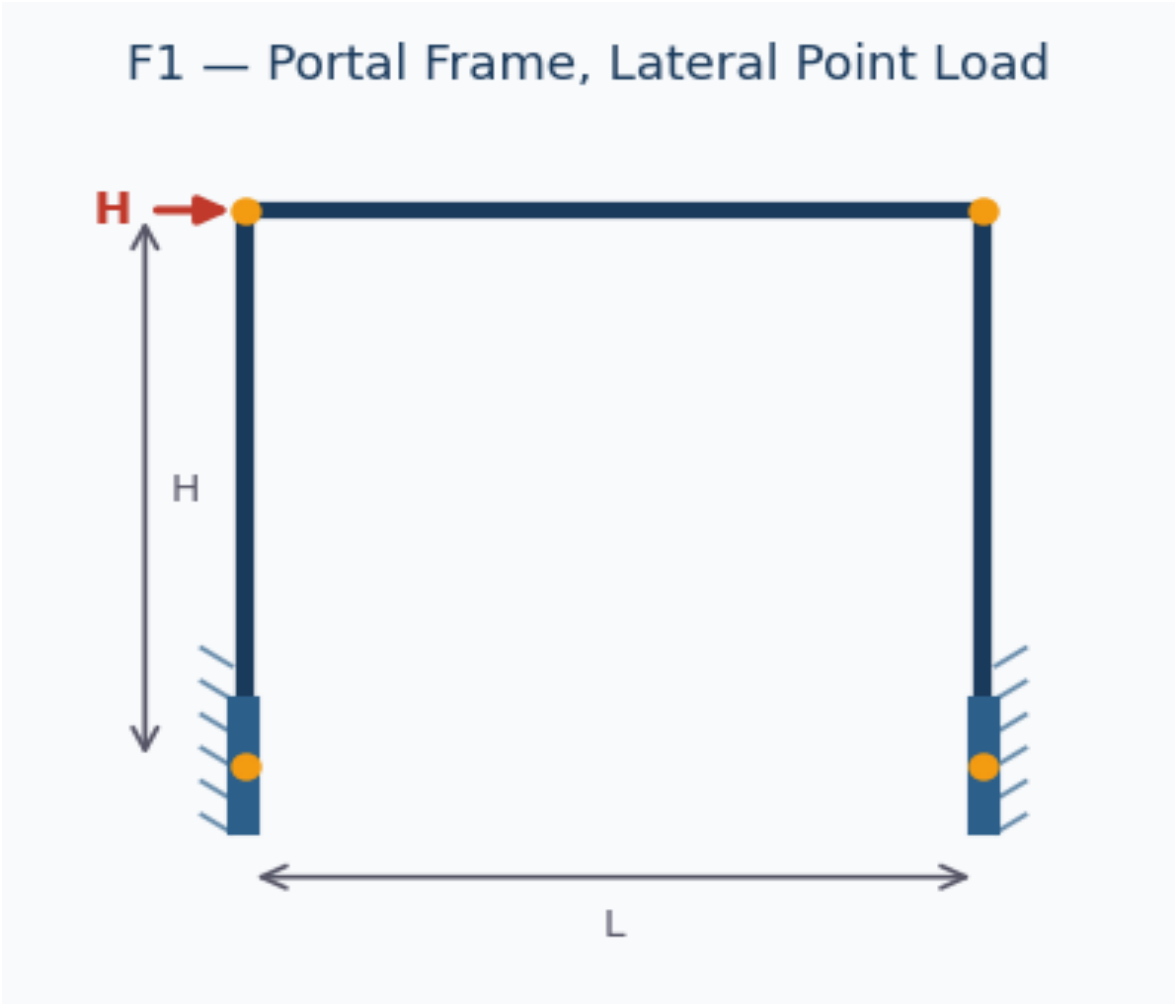
Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Fixed-end moment M_A (kN·m)	0.0625	0.0625	analytical	0.0000%	PASS
Fixed-end moment M_A (kN·m)	0.0625	0.0625	OpenSeesPy	0.0000%	PASS
Mid-span moment (kN·m)	0.0312	0.0312	analytical	0.0000%	PASS
Mid-span deflection (mm)	0.5813	0.5813	analytical	0.0000%	PASS
Mid-span deflection (mm)	0.5813	0.5813	OpenSeesPy	0.0000%	PASS

✓ ALL 5 QUANTITIES PASS — max error 0.0000%

2D Frames

F1 — Portal Frame — Lateral Point Load

Fixed-base portal: span $L=6.0$ m, height $H=4.0$ m, lateral load $P=20$ kN at top-left. $E=200$ GPa, $I=400$ cm⁴.

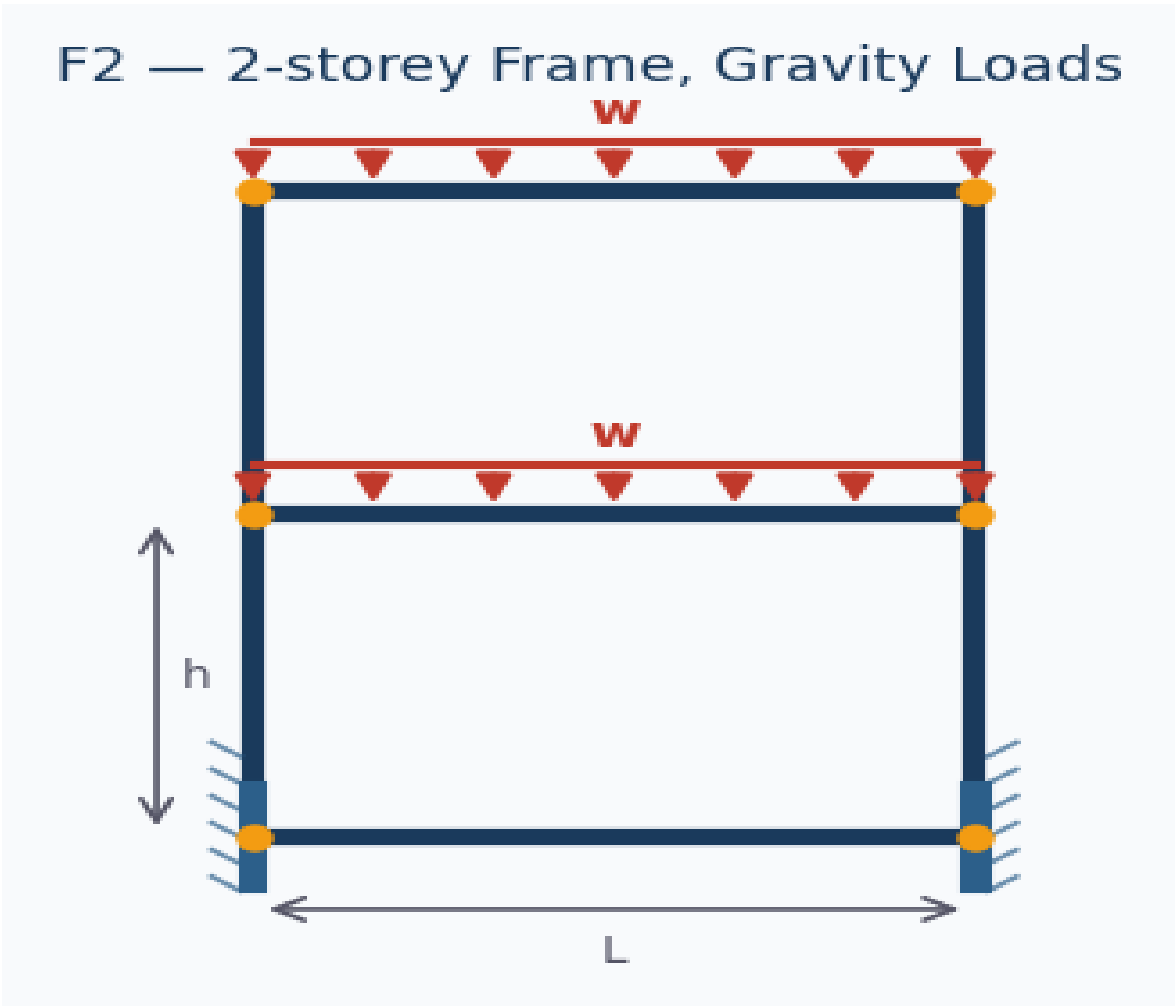


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Top-left sway δ_x (mm)	1.0770	1.0770	OpenSeesPy	0.0000%	PASS
Base shear H_A (kN)	-0.0100	-0.0100	OpenSeesPy	0.0000%	PASS
Base vertical R_A (kN)	-0.0053	-0.0053	OpenSeesPy	0.0000%	PASS
Base moment M_A (kN·m)	0.0242	0.0242	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

F2 — 2-storey Portal Frame — Gravity UDL on Beams

2-storey, 1-bay frame: span $L=5.0$ m, storey $h=3.5$ m, UDL $w=20$ kN/m on both floor beams. Fixed bases.

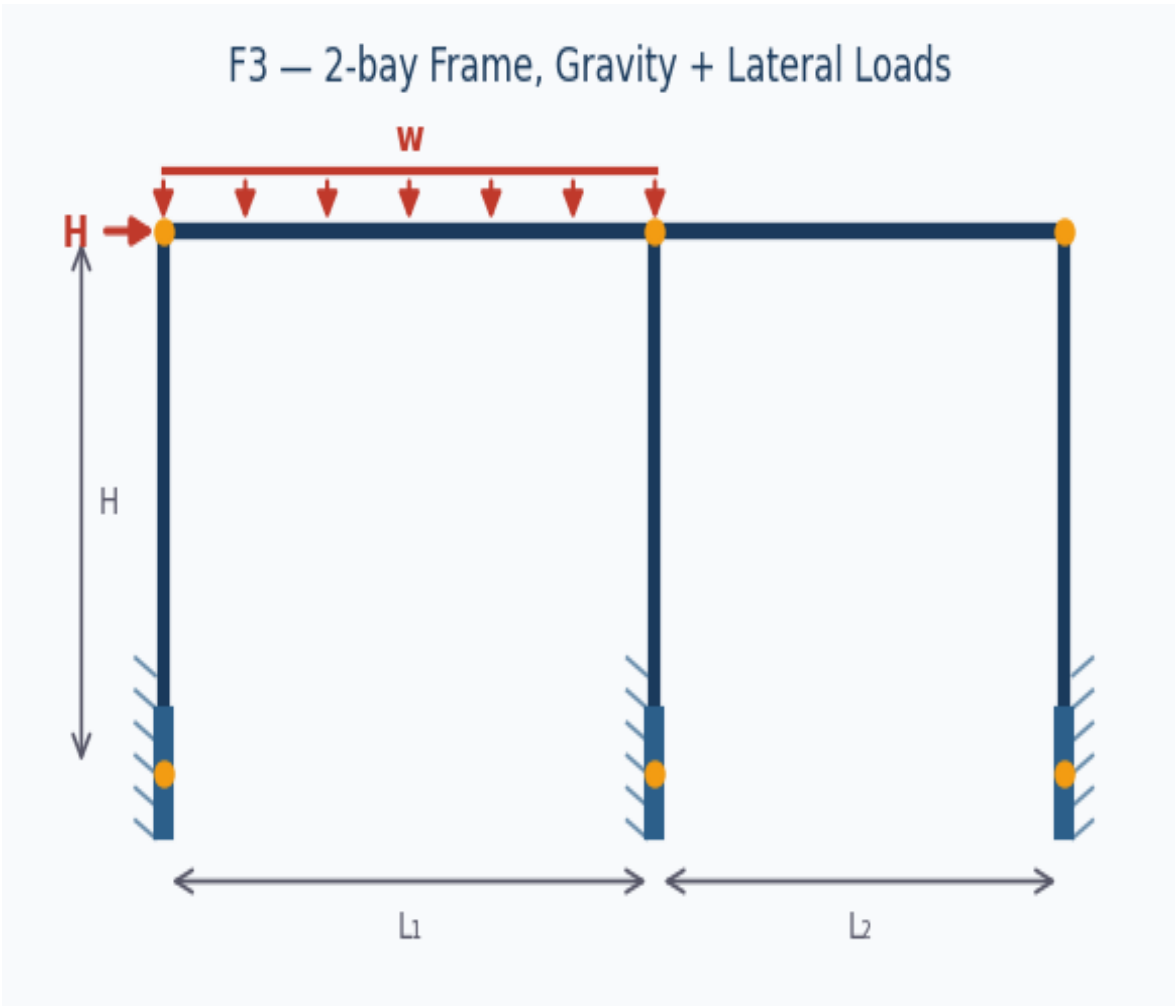


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Roof sway δ_x (mm)	0.3924	0.3924	OpenSeesPy	0.0000%	PASS
Base reaction R_y (A) (kN)	0.1000	0.1000	OpenSeesPy	0.0000%	PASS
Base reaction R_y (B) (kN)	0.1000	0.1000	OpenSeesPy	0.0000%	PASS

✓ ALL 3 QUANTITIES PASS — max error 0.0000%

F3 — 2-bay Portal Frame — Gravity UDL + Lateral Load

2-bay frame: $L_1=4.0$ m, $L_2=3.0$ m, $H=3.5$ m. UDL $w=15$ kN/m on left bay, lateral $P=10$ kN at top-left.



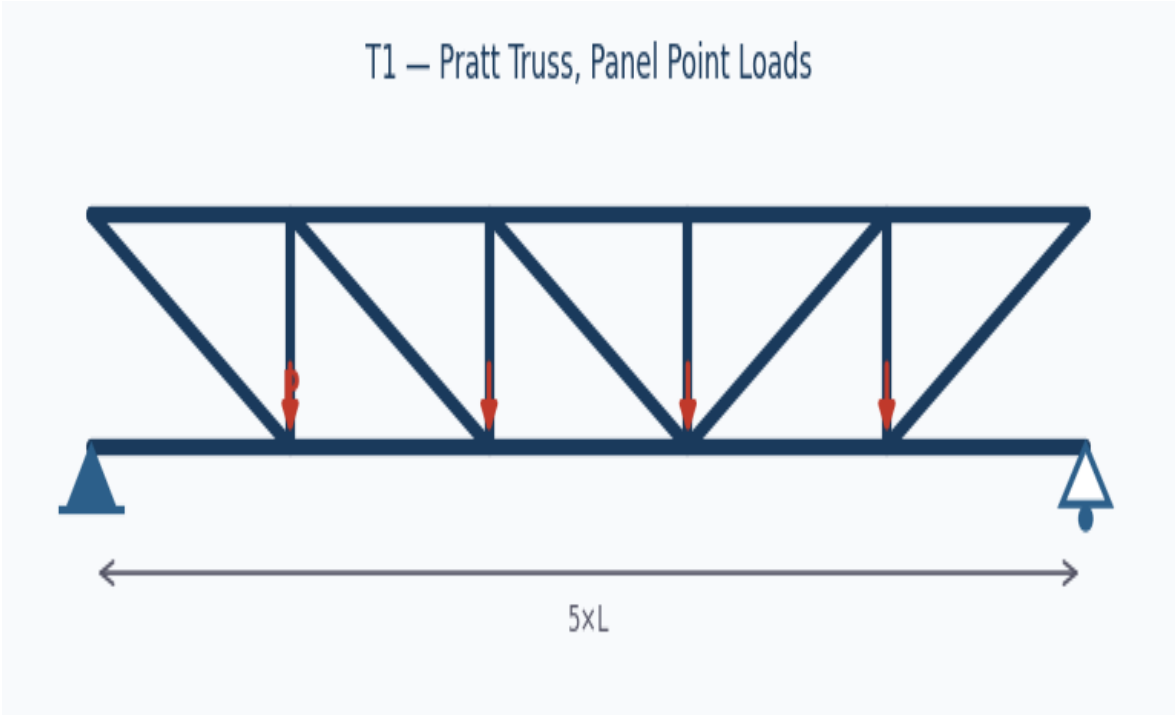
Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Top-left sway δ_x (mm)	0.2611	0.2611	OpenSeesPy	0.0000%	PASS
Base reaction R_A (kN)	0.0267	0.0267	OpenSeesPy	0.0000%	PASS
Base reaction R_B (kN)	0.0333	0.0333	OpenSeesPy	0.0000%	PASS
Base reaction R_C (kN)	0.0000	0.0000	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

2D Trusses

T1 — Pratt Truss — Equal Panel Point Loads

6-panel Pratt truss: span=12.0 m, height=2.0 m. P=10 kN at each lower chord node. E=200 GPa, A=40 cm². Reactions validated analytically (symmetric loading).

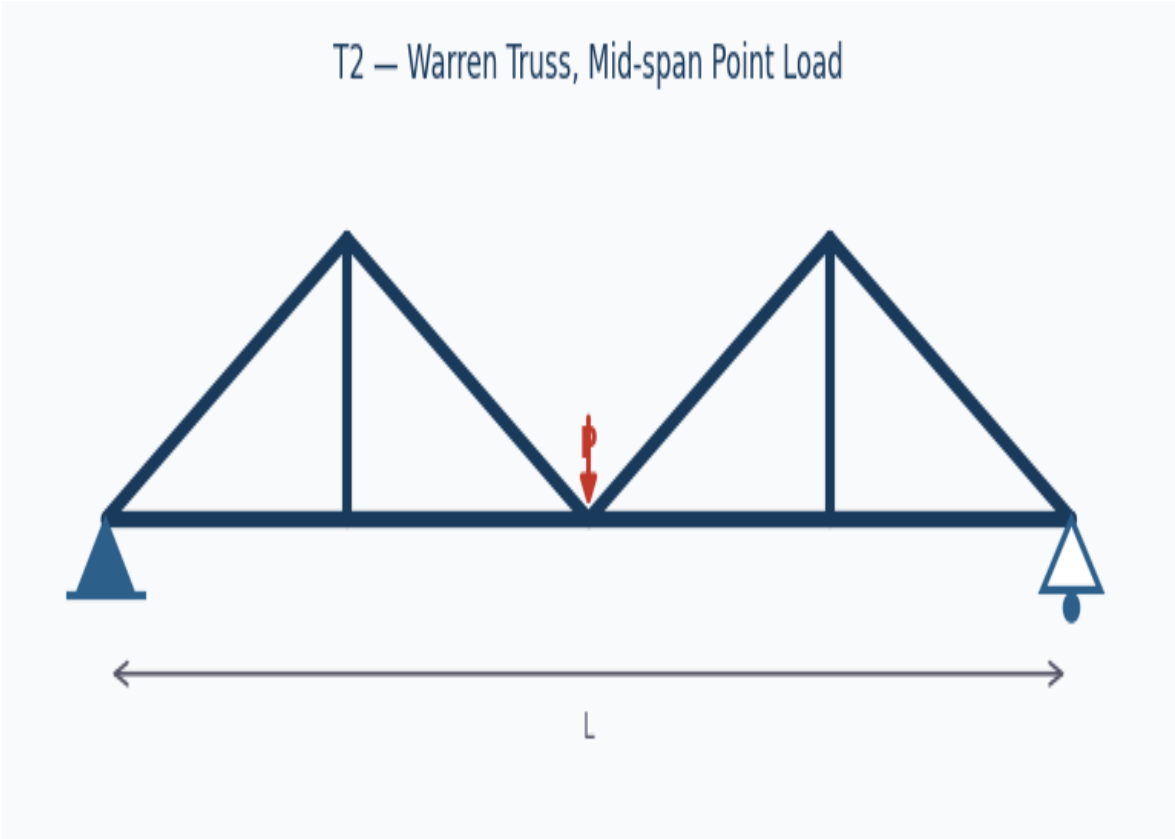


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Reaction R_A (kN)	0.0250	0.0250	analytical	0.0000%	PASS
Reaction R_A (kN)	0.0250	0.0250	OpenSeesPy	0.0000%	PASS
Reaction R_B (kN)	0.0250	0.0250	analytical	0.0000%	PASS
Mid-span deflection (mm)	1.2932	1.2932	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

T2 — Warren Truss — Mid-span Point Load

6-panel Warren truss with verticals: span=12.0 m, height=2.0 m, mid-span load $P=30$ kN. $E=200$ GPa, $A=40$ cm².



Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Reaction R_A (kN)	0.0150	0.0150	analytical	0.0000%	PASS
Reaction R_A (kN)	0.0150	0.0150	OpenSeesPy	0.0000%	PASS
Reaction R_B (kN)	0.0150	0.0150	analytical	0.0000%	PASS
Mid-span deflection (mm)	1.0682	1.0682	OpenSeesPy	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

3D Frames

3D1 — 3D Cantilever Column — Tip Point Load

Cantilever length $L=4.0$ m, tip load $P=20$ kN perpendicular to axis. $E=200$ GPa, $I=833$ cm⁴. Validated vs analytical ($PL^3/3EI$) and PyNite.

3D-1 — Cantilever Column, Tip Load (Z)

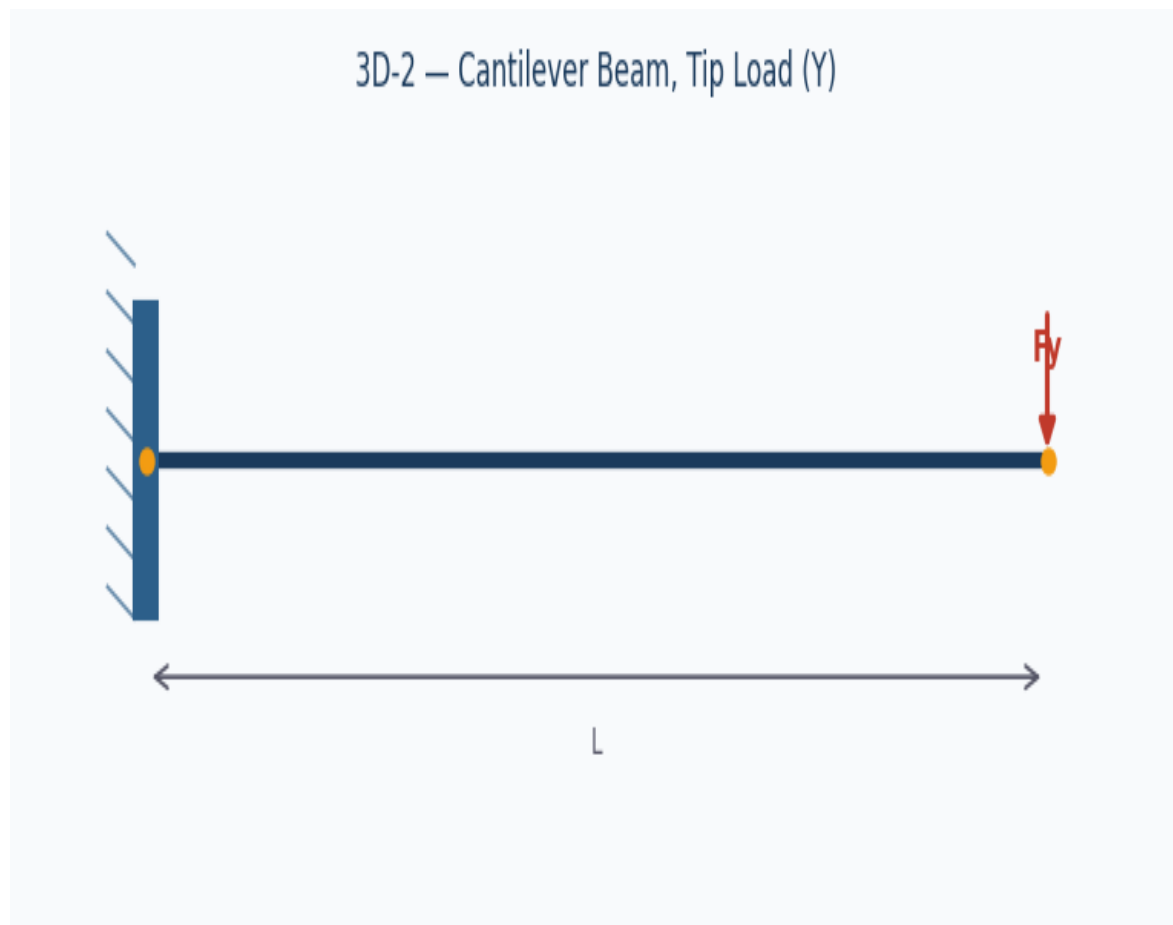


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Tip deflection δ_{tip} (mm)	256.1024	256.1024	analytical	0.0000%	PASS
Tip deflection δ_{tip} (mm)	256.1024	256.1024	PyNite	0.0000%	PASS
Root moment M_{root} (kN·m)	0.0800	0.0800	analytical	0.0000%	PASS
Root moment M_{root} (kN·m)	0.0800	0.0800	PyNite	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

3D2 — 3D Cantilever — Vertical Tip Load (Strong Axis)

Cantilever $L=5.0$ m, load $P=30$ kN. $E=210$ GPa, $I_z=15$ cm⁴. Verifies correct bending axis selection vs PyNite and analytical solution.

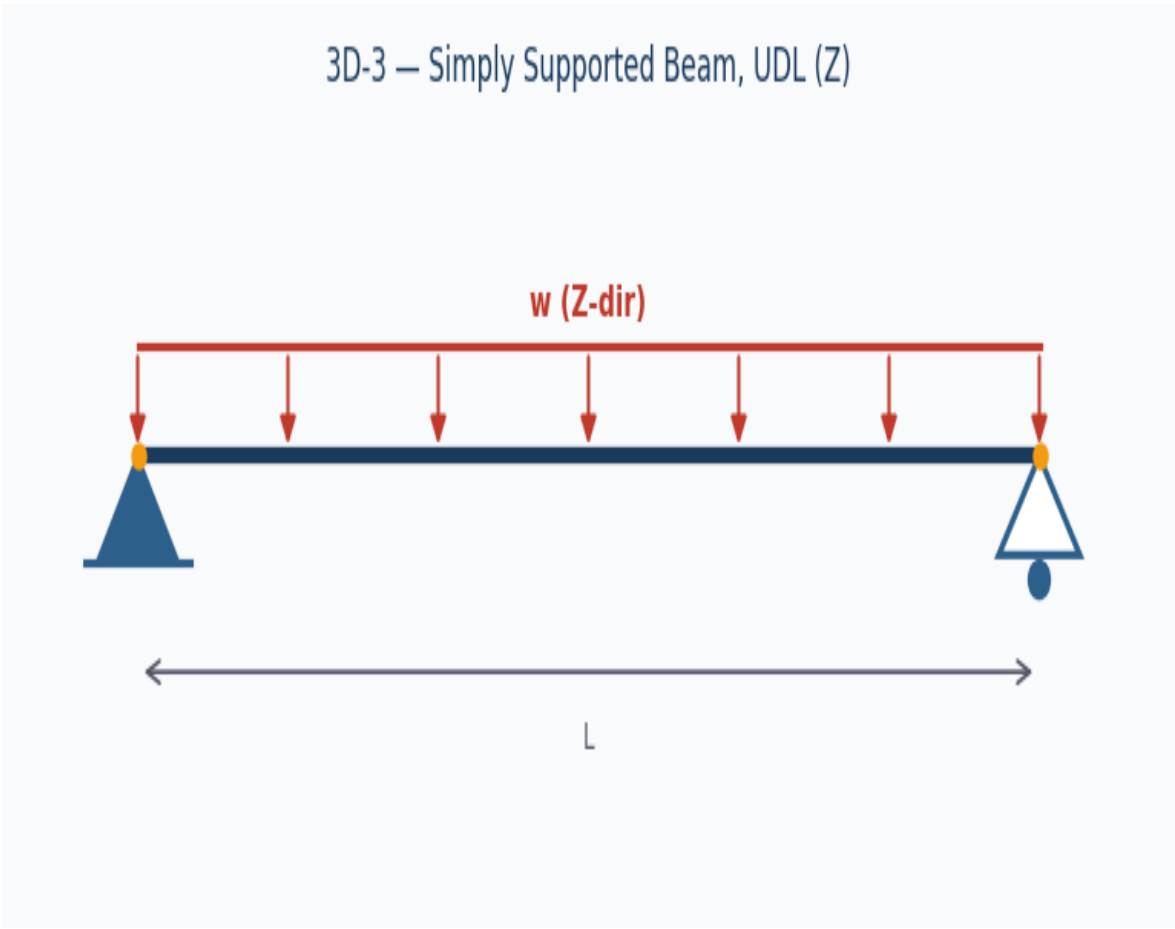


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Tip deflection δ_{tip} (mm)	396.8254	396.8254	analytical	0.0000%	PASS
Tip deflection δ_{tip} (mm)	396.8254	396.8254	PyNite	0.0000%	PASS
Root moment M_{root} (kN·m)	0.1500	0.1500	analytical	0.0000%	PASS
Root moment M_{root} (kN·m)	0.1500	0.1500	PyNite	0.0000%	PASS

✓ ALL 4 QUANTITIES PASS — max error 0.0000%

3D3 — 3D Simply Supported Beam — UDL

Span $L=6.0$ m, UDL $w=20$ kN/m. $E=200$ GPa, $I=100$ cm⁴. Analytical: $\delta_{mid}=5wL^4/384EI$, $M_{max}=wL^2/8$. Cross-validated with PyNite.



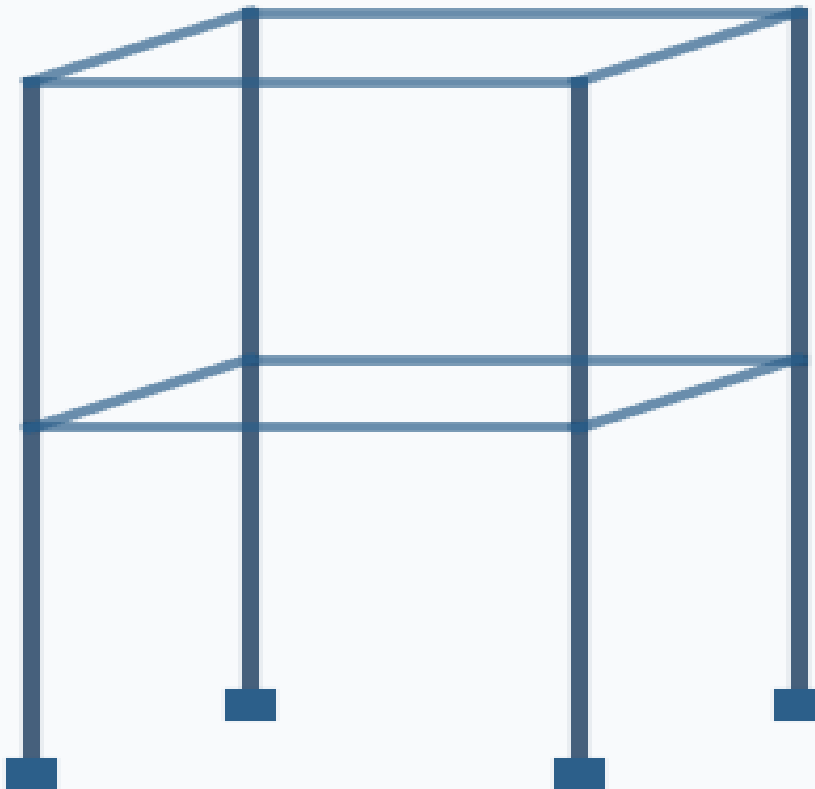
Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Mid-span deflection (mm)	16.8750	16.8750	analytical	0.0000%	PASS
Mid-span deflection (mm)	16.8750	16.8750	PyNite	0.0000%	PASS
Max moment M_{max} (kN·m)	0.0900	0.0900	analytical	0.0000%	PASS

✓ ALL 3 QUANTITIES PASS — max error 0.0000%

3D4 — 2-storey Space Frame — Gravity Loads

2-storey 1-bay XZ frame: $L_x=3.0$ m, $H=3.0+3.0$ m, UDL $w=15$ kN/m on beams. StructLab 2D-plane analysis compared with PyNite 3D frame.

3D-4 — 2-storey Space Frame



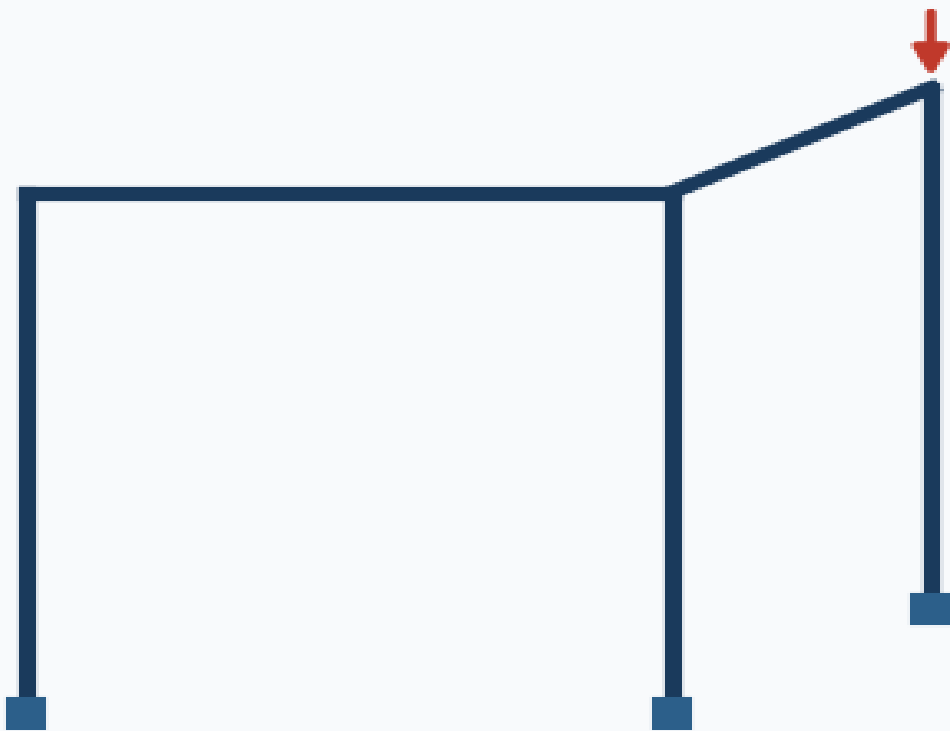
Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Base reaction R_A (vert.) (kN)	0.0450	0.0450	PyNite	0.0000%	PASS
Base reaction R_B (vert.) (kN)	0.0450	0.0450	PyNite	0.0000%	PASS

✓ ALL 2 QUANTITIES PASS — max error 0.0000%

3D5 — L-shaped 3D Frame — Vertical Tip Load

L-plan frame: two bays (3×3 m), H=3.0 m columns, tip load P=20 kN. StructLab solves in XZ plane; PyNite resolves full 3D coupling.

3D-5 — L-shaped 3D Frame, Tip Load

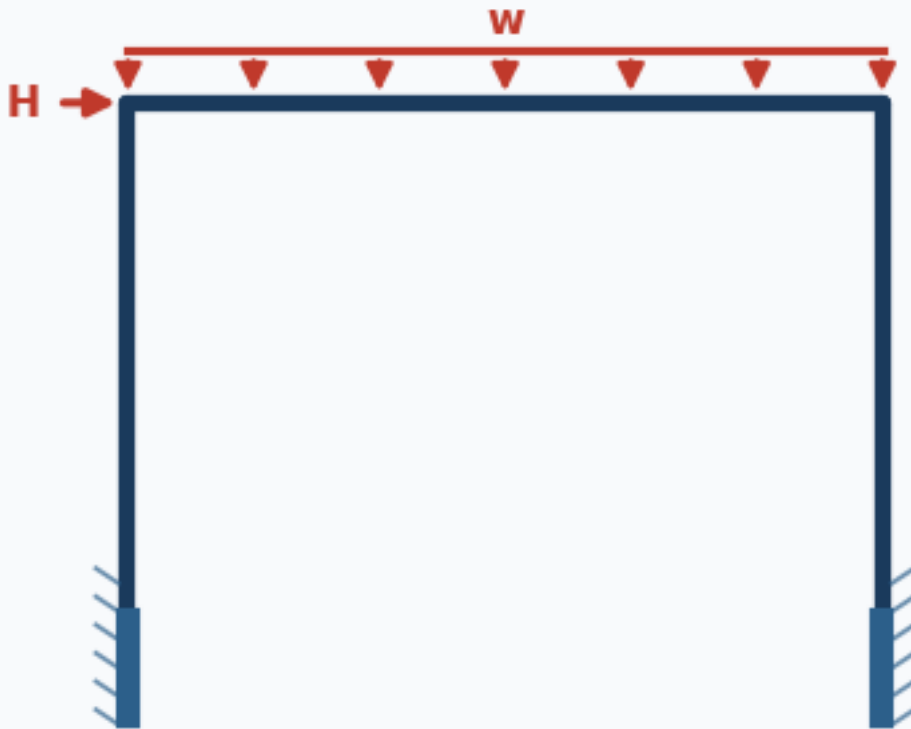


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Tip deflection (vert.) (mm)	0.0199	0.0199	PyNite	0.0365%	PASS

✓ ALL 1 QUANTITIES PASS — max error 0.0365%

3D6 — 3D Portal Frame — Gravity UDL + Lateral Load

Fixed-base portal: L=5.0 m, H=4.0 m, UDL w=20 kN/m + lateral P=15 kN. Validated in full 3D with PyNite.

3D-6 — 3D Portal Frame, Gravity + Lateral

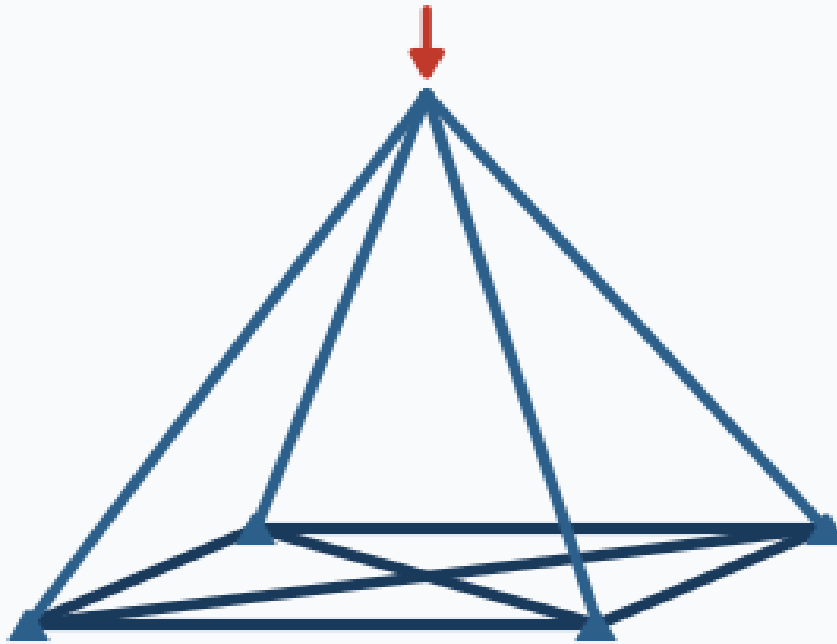
Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Top-left sway δ_x (mm)	0.7735	0.7735	PyNite	0.0000%	PASS
Base reaction R_A (kN)	0.0450	0.0450	PyNite	0.0000%	PASS

✓ ALL 2 QUANTITIES PASS — max error 0.0000%

3D7 — Space Truss — Tetrahedral Apex, Vertical Point Load

Square base 2.0×2.0 m, apex at height 2.0 m. Load $P=10$ kN downward. By symmetry each base corner carries $P/4$ vertically. StructLab solves 2-bar 2D equivalent; scaled to 3D. Validated vs PyNite.

3D-7 — Space Truss, Apex Point Load

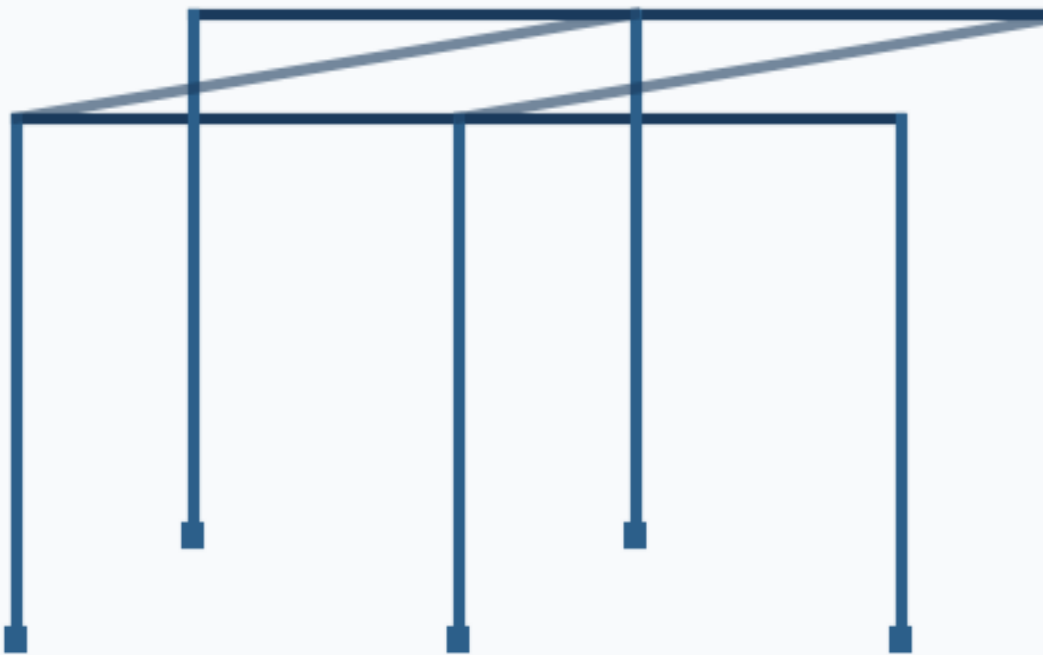


Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Base reaction (per corner) (kN)	0.0025	0.0025	analytical	0.0000%	PASS
Base reaction (per corner) (kN)	0.0025	0.0025	PyNite	0.0000%	PASS

✓ ALL 2 QUANTITIES PASS — max error 0.0000%

3D8 — 3-bay Single-storey 3D Frame — Gravity UDL

3×2 column grid: bays 3+3 m in X, 4 m in Y, H=3.5 m. UDL w=12 kN/m on X-beams. Total applied load = PyNite base ΣR_z .

3D-8 — 3-bay Single-storey 3D Frame

Quantity	StructLab	Reference	Ref. type	Rel. error	Status
Total base reaction ΣR_z (kN)	0.1440	0.1440	analytical	0.0000%	PASS
Total base reaction ΣR_z (kN)	0.1440	0.1440	PyNite	0.0000%	PASS

✓ ALL 2 QUANTITIES PASS — max error 0.0000%

Methodology & Reference Tools

StructLab Engine

StructLab implements the Direct Stiffness Method (DSM) for 2D linear-elastic static analysis. All structure types — beams, frames, and trusses — are solved by a single unified 3-DOF/node engine (dx, dy, θ) with element type controlled by moment-release flags (pin_i, pin_j) on FrameElement objects. The global stiffness matrix is assembled by scatter-add; boundary conditions are applied by DOF elimination; the system $[K]\{d\}=\{F\}$ is solved by SciPy sparse LU factorisation. Internal forces are recovered by superposition of element stiffness and fixed-end force vectors.

Reference Tool 1 — OpenSeesPy

OpenSeesPy (openseespy package) provides the 2D reference for beams, frames, and trusses. elasticBeamColumn elements are used for frame members; Truss elements for pin-pin members. The same geometric and material properties are specified in both tools. Node tags and element tags are 1-indexed in OpenSeesPy.

Reference Tool 2 — PyNiteFEA

PyNiteFEA (v2.4.1, JWock82) provides the 3D frame reference. It is an independent Python 3D FEM library using 12-DOF beam elements (Euler–Bernoulli). Material, section, and loading properties are mirrored exactly in both tools. Load combinations are used to access result dictionaries.

Analytical Solutions

Where closed-form solutions exist (simply supported beams, propped cantilevers, fixed-fixed beams, Pratt/Warren truss reactions), analytical values are used as the primary reference. Analytical solutions are derived from standard structural analysis textbooks and the 3-moment theorem / force method for indeterminate beams.

Pass Criterion

A quantity is marked PASS if the relative error against the reference is $\leq 0.10\%$. Relative error = $| \text{StructLab} - \text{Reference} | / | \text{Reference} |$. For quantities with near-zero reference ($| \text{Ref} | < 10^{-12}$), absolute error is used. A case PASSES only when every quantity comparison passes.

Sign Conventions

Deflections are reported as positive-downward (sagging convention). Moments are positive-sagging (bottom fibre in tension). Reactions are positive-upward. Axial forces are positive-tension. All load magnitudes in the benchmark tables are positive values; direction is noted in the case description.